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A. Untuk nomor 1 sampai dengan nomor 4, tentukan Penyelesaian (Umum atau khusus) Persamaan Diferensial Linear berikut :

1.) $\frac{dy}{dx} - y = x$

$P = -1 \quad , \quad Q = x$

$y = e^{-\int P dx} \left(\int Q e^{\int P dx} \, dx + c \right)$

$y = e^{-\int -1 \, dx} \left(\int x e^{\int -1 \, dx} \, dx + c \right)$

*hitung satu persatu :

$\cdot\rangle\ e^{-\int -1 \, dx} = e^{-(-x)} = e^{x}$

$\cdot\rangle\ \int x e^{\int -1 \, dx} \, dx = \int x e^{-x} \, dx$

$u = x \quad , \quad dv = e^{-x} dx \quad , \quad du = dx \quad , \quad v = -e^{-x}$

$\int u dv = uv - \int v du$

$\int x e^{-x} dx = -x e^{-x} - \int -e^{-x} \, dx$

$= -x e^{-x} + \int e^{-x} \, dx$

$= -x e^{-x} + (-e^{-x})$

$= -x e^{-x} - e^{-x}$

$y = e^{x} (-x e^{-x} - e^{-x} + c)$

$y = -x - 1 + c e^{x}$ → solusi umum

2.) $\frac{dy}{dx} - 2y = 8e^{x}$

$P = -2 \quad , \quad Q = 8e^{x}$

$y = e^{-\int P dx} \left(\int Q e^{\int P dx} \, dx + c \right)$

$y = e^{-\int -2 \, dx} \left(\int 8e^{x} \cdot e^{\int -2 \, dx} \, dx + c \right)$

$y = e^{-(-2x)} \left(8 \int e^{x} \cdot e^{-2x} \, dx + c \right)$

$y = e^{2x} \left(8 \int e^{-x} \, dx + c \right)$

$y = e^{2x} \left(8 (-e^{-x}) + c \right)$

$y = e^{2x} (-8e^{-x} + c)$

$y = -8e^{x} + c e^{2x}$ → solusi umum

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3.) $3\frac{dy}{dx} + 5y = 20, y(0) = 2$

$\frac{dy}{dx} + \frac{5}{3}y = \frac{20}{3}, y(0) = 2$

$P = \frac{5}{3}, Q = \frac{20}{3}$

$y = e^{-\int P dx}\left(\int Q e^{\int P dx} dx + c\right)$

$y = e^{-\int 5/3\, dx}\left(\int \frac{20}{3} e^{\int 5/3\, dx} dx + c\right)$

$y = e^{-5/3x}\left(\int \frac{20}{3} e^{5/3x} dx + c\right)$

$y = e^{-5/3x}\left(\frac{20}{3}\int e^{5/3x} dx + c\right)$

$y = e^{-5/3x}\left(\frac{20}{3}\left(\frac{3}{5}e^{5/3x}\right) + c\right)$

$y = e^{-5/3x}\left(4e^{5/3x} + c\right)$

$y = 4 + ce^{-5/3x}$ → solusi umum

$y(0) = 2 \rightarrow 2 = 4 + ce^{-5/3(0)}$

$2 = 4 + ce^{0}$

$2 = 4 + c(1) \rightarrow c = -2$

$y = 4 + -2e^{-5/3x}$ → solusi khusus

4) $2\frac{dy}{dx} + 3y = 5\sin x$, dengan $x = 0, y = 2$

$\frac{dy}{dx} + \frac{3}{2}y = \frac{5}{2}\sin x$, dengan $x = 0, y = 2$

$P = \frac{3}{2}, Q = \frac{5}{2}\sin x$

$y = e^{-\int P dx}\left(\int Q e^{\int P dx} dx + c\right)$

$y = e^{-\int 3/2\, dx}\left(\int \frac{5}{2}\sin x \cdot e^{\int 3/2\, dx} dx + c\right)$

$y = e^{-3/2x}\left(\frac{5}{2}\int e^{3/2x}\sin x\, dx + c\right)$

* hitung integral $\int e^{3/2x}\sin x$

$\int e^{3/2x}\sin x\, dx$

$u = \sin x, dv = e^{3/2x}dx, du = \cos x\, dx, v = \frac{2}{3}e^{3/2x}$

$\int dvu = vu - \int v\, du$

$\int e^{3/2x}\cdot\sin x\, dx = \frac{2}{3}e^{3/2x}\sin x - \int \frac{2}{3}e^{3/2x}\cdot\cos x\, dx$

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$\int e^{3/2x} \sin x \, dx = \frac{2}{3} e^{3/2x} \sin x - \frac{2}{3} \int e^{3/2x} \cos x \, dx$

* hitung integral $\int e^{3/2x} \cos x \, dx$

$u = \cos x \quad dv = e^{3/2x} dx \quad du = -\sin x \, dx \quad v = \frac{2}{3} e^{3/2x}$

$\int dv\, u = vu - \int v \, du$

$\int e^{3/2x} \cos x \, dx = \frac{2}{3} e^{3/2x} \cos x - \int \frac{2}{3} e^{3/2x} (-\sin x \, dx)$

$\int e^{3/2x} \cos x \, dx = \frac{2}{3} e^{3/2x} \cos x + \frac{2}{3} \int e^{3/2x} \sin x \, dx$

* Substitusi hasil integral $\int e^{3/2x} \cos x \, dx$ ke persamaan integral $\int e^{3/2x} \sin x \, dx$

$\int e^{3/2x} \sin x \, dx = \frac{2}{3} e^{3/2x} \sin x - \frac{2}{3} \int e^{3/2x} \cos x \, dx$

$\int e^{3/2x} \sin x \, dx = \frac{2}{3} e^{3/2x} \sin x - \frac{2}{3} \left(\frac{2}{3} e^{3/2x} \cos x + \frac{2}{3} \int e^{3/2x} \sin x \, dx \right)$

$\int e^{3/2x} \sin x \, dx = \frac{2}{3} e^{3/2x} \sin x - \frac{4}{9} e^{3/2x} \cos x - \frac{4}{9} \int e^{3/2x} \sin x \, dx$

$\int e^{3/2x} \sin x \, dx + \frac{4}{9} \int e^{3/2x} \sin x \, dx = \frac{2}{3} e^{3/2x} \sin x - \frac{4}{9} e^{3/2x} \cos x$

$\frac{9}{9} \int e^{3/2x} \sin x \, dx + \frac{4}{9} \int e^{3/2x} \sin x \, dx = \frac{2}{3} e^{3/2x} \sin x - \frac{4}{9} e^{3/2x} \cos x$

$\frac{13}{9} \int e^{3/2x} \sin x \, dx = \frac{2}{3} e^{3/2x} \sin x - \frac{4}{9} e^{3/2x} \cos x$

$\int e^{3/2x} \sin x \, dx = \frac{9}{13} \left(\frac{2}{3} e^{3/2x} \sin x - \frac{4}{9} e^{3/2x} \cos x \right)$

$\int e^{3/2x} \sin x \, dx = \frac{6}{13} e^{3/2x} \sin x - \frac{4}{13} e^{3/2x} \cos x$

* Substitusi hasil integral $\int e^{3/2x} \sin x \, dx$ ke persamaan y

$y = e^{-3/2x} \left(\frac{5}{2} \int e^{3/2x} \sin x \, dx + c \right)$

$y = e^{-3/2x} \left(\frac{5}{2} \left(\frac{6}{13} e^{3/2x} \sin x - \frac{4}{13} e^{3/2x} \cos x \right) + c \right)$

$y = e^{-3/2x} \left(\frac{15}{13} e^{3/2x} \sin x - \frac{10}{13} e^{3/2x} \cos x + c \right)$

$y = \frac{15}{13} \sin x - \frac{10}{13} \cos x + c e^{-3/2x}$ → solusi umum

$x = 0, y = 2 \rightarrow 2 = \frac{15}{13} \sin 0 - \frac{10}{13} \cos 0 + c e^{-3/2(0)}$

$2 = 0 - \frac{10}{13} + c$

$2 + \frac{10}{13} = c$

$\frac{26}{13} + \frac{10}{13} = c$

$c = \frac{36}{13}$

$y = \frac{15}{13} \sin x - \frac{10}{13} \cos x + \frac{36}{13} e^{-3/2x}$ → solusi khusus

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B. 1) Rangkaian listrik RL seri dengan tahanan $R = 20\ \Omega$, induktansi $L = 2$ henry yang dihubungkan dengan tegangan sumber $E(t) = 12$ volt

a. Tentukan persamaan diferensial hubungan input $E(t)$ dan output $i(t)$

$$L\frac{di}{dt} + Ri = V$$

$$2\frac{di}{dt} + 20i = 12$$

$$\frac{di}{dt} + 10i = 6$$

b. Tentukan solusi /output yaitu $i(t)$ bila $i(t=0) = 0$

Faktor integrasi $M = e^{\int 10\,dt} = e^{10t}$

$$i(t) = \frac{\int 6e^{10t}\,dt}{e^{10t}}$$

$$i(t) = \frac{\frac{6}{10}e^{10t} + C}{e^{10t}}$$

$$i(t) = e^{-10t}\left(\frac{6}{10}e^{10t} + C\right)$$

$$i(t) = \frac{6}{10} + Ce^{-10t} \rightarrow \text{solusi umum}$$

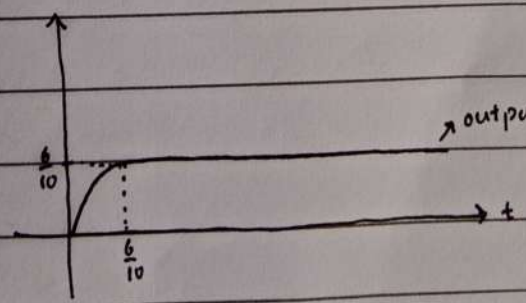
bila $i(t=0) = 0 \rightarrow 0 = \frac{6}{10} + Ce^{-10(0)}$

$$0 = \frac{6}{10} + C$$

$$C = -\frac{6}{10}$$

$$i(t) = \frac{6}{10} - \frac{6}{10}e^{-10t} \rightarrow \text{solusi khusus}$$

c. Gambarkan grafik input dan output



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2) Rangkaian listrik RL seri dengan tahanan $R = 20\ \Omega$, induktansi $L = 2$ henry yang dihubungkan dengan tegangan sumber $E(t) = 12 \sin 3t$ volt

a. Tentukan persamaan diferensial hubungan input $E(t)$ dan output $i(t)$

$L \frac{di}{dt} + Ri = V$

$2 \frac{di}{dt} + 20i = 12 \sin 3t$

$\frac{di}{dt} + 10i = 6 \sin 3t$

b. Tentukan solusi/output yaitu arus $i(t)$ bila $i(t=0) = 0$

Faktor integrasi $\mu = e^{\int 10\, dt} = e^{10t}$

$i(t) = \dfrac{\int 6 \sin 3t \cdot e^{10t}\, dt}{e^{10t}}$

$i(t) = \dfrac{6 \int e^{10t} \sin 3t\, dt}{e^{10t}}$

* hitung integral $\int e^{10t} \sin 3t\, dt$

$u = \sin 3t \quad dv = e^{10t} dt \quad du = 3 \cos 3t\, dt \quad v = \frac{1}{10} e^{10t}$

$\int dvu = vu - \int v\, du$

$\int e^{10t} \sin 3t = \frac{1}{10} e^{10t} \sin 3t - \int \frac{1}{10} e^{10t} (3 \cos 3t\, dt)$

$= \frac{1}{10} e^{10t} \sin 3t - \frac{3}{10} \int e^{10t} \cos 3t\, dt$

* hitung integral $\int e^{10t} \cos 3t\, dt$

$u = \cos 3t \quad dv = e^{10t} dt \quad du = -3 \sin 3t\, dt \quad v = \frac{1}{10} e^{10t}$

$\int dvu = vu - \int v\, du$

$\int e^{10t} \cos 3t\, dt = \frac{1}{10} e^{10t} \cos 3t - \int \frac{1}{10} e^{10t} (-3 \sin 3t\, dt)$

$= \frac{1}{10} e^{10t} \cos 3t + \frac{3}{10} \int e^{10t} \sin 3t\, dt$

* Substitusi hasil integral $\int e^{10t} \cos 3t\, dt$ ke persamaan $\int e^{10t} \sin 3t$

$\int e^{10t} \sin 3t\, dt = \frac{1}{10} e^{10t} \sin 3t - \frac{3}{10} \int e^{10t} \cos 3t\, dt$

$\int e^{10t} \sin 3t\, dt = \frac{1}{10} e^{10t} \sin 3t - \frac{3}{10} \left(\frac{1}{10} e^{10t} \cos 3t + \frac{3}{10} \int e^{10t} \sin 3t\, dt \right)$

$\int e^{10t} \sin 3t\, dt = \frac{1}{10} e^{10t} \sin 3t - \frac{3}{100} e^{10t} \cos 3t - \frac{9}{100} \int e^{10t} \sin 3t\, dt$

$\int e^{10t} \sin 3t\, dt + \frac{9}{100} \int e^{10t} \sin 3t\, dt = \frac{1}{10} e^{10t} \sin 3t - \frac{3}{100} e^{10t} \cos 3t$

$\frac{109}{100} \int e^{10t} \sin 3t\, dt = \frac{1}{10} e^{10t} \sin 3t - \frac{3}{100} e^{10t} \cos 3t$

$\int e^{10t} \sin 3t \, dt = \frac{100}{109}\left(\frac{1}{10} e^{10t} \sin 3t - \frac{3}{100} e^{10t} \cos 3t\right)$

$= \frac{10}{109} e^{10t} \sin 3t - \frac{3}{109} e^{10t} \cos 3t$

* substitusi hasil integral $\int e^{10t} \sin 3t \, dt$ ke persamaan i(t)

$i(t) = 6e^{-10t} \int e^{10t} \sin 3t \, dt$

$i(t) = 6e^{-10t}\left(\left(\frac{10}{109} e^{10t} \sin 3t - \frac{3}{109} e^{10t} \cos 3t\right) + c\right)$

$i(t) = \frac{60}{109} \sin 3t - \frac{18}{109} \cos 3t + ce^{-10t}$ → solusi umum

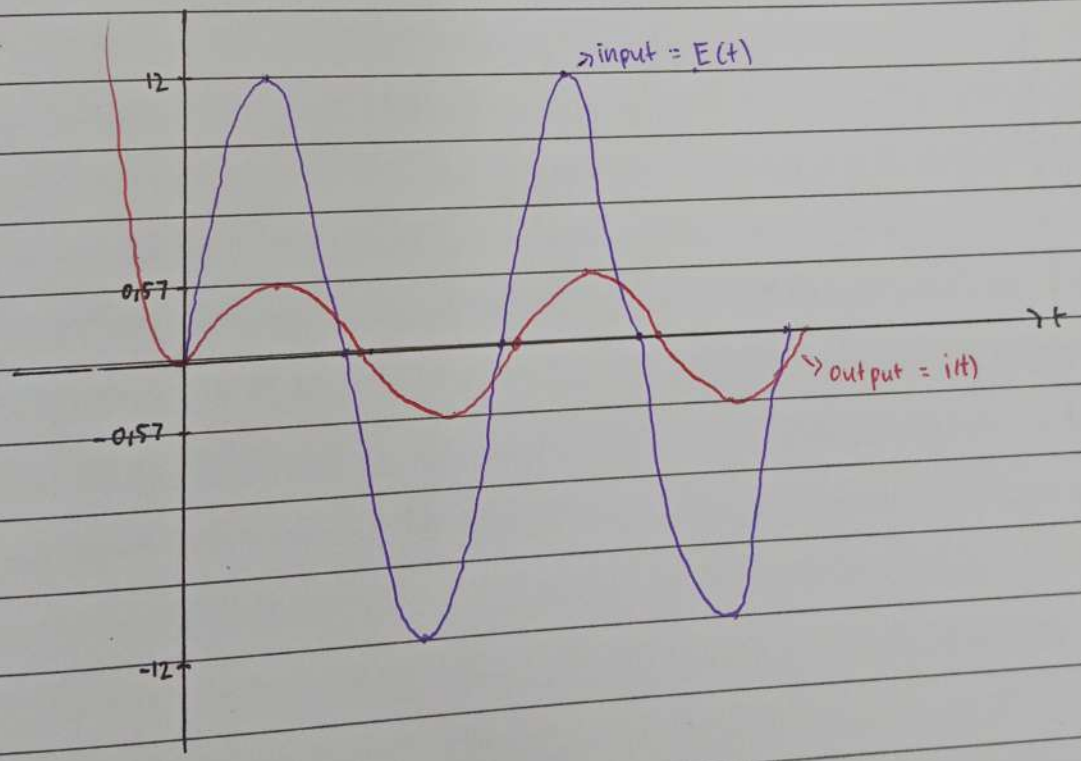
bila $i(t=0) = 0$ → $0 = \frac{60}{109} \sin 3(0) - \frac{18}{109} \cos 3(0) + ce^{-10(0)}$

$0 = 0 - \frac{18}{109} \cdot 1 + c \cdot 1$

$c = \frac{18}{109}$

$i(t) = \frac{60}{109} \sin 3t - \frac{18}{109} \cos 3t + \frac{18}{109} e^{-10t}$ → solusi khusus

c. Gambarkan grafik input dan output



Contoh 2 :

Rangkaian listrik RL dengan emf V = 5 volt, R = 50 ohm, dan L = 1 H, dengan rumus $L\frac{di}{dt} + Ri = V$. Tentukan arus i(t) untuk t > 0 bila pada t = 0, i = 0

Penyelesaian :

Model matematika pada contoh 2 untuk V = 5 volt, R = 50 ohm, dan L = 1 adalah $\frac{di}{dt} + 50i = 5$

Faktor integral $e^{\int 50\,dt} = 50^{50t}$ dan $ie^{50t} = 5\int e^{50t}\,dt = \frac{1}{10}e^{50t} + k$

Pada t = 0, i = 0 diperoleh k = -a1

Jadi penyelesaian PD adalah $i(t) = 0{,}1\,(1 - e^{-50t})$

Arus transien (transient current) $i(t) = 0{,}1\,(1 - e^{-50t})$ dan arus mantap (steady state current)

i = 0,1 ampere dengan grafik seperti pada gambar 1. Geogebra

contoh 3 :

Tentukan muatan untuk t > 0 dalam rangkaian listrik seri RC dengan R = 10 ohm, C = $4 \cdot 10^{-3}$ F dan sumber V = 85 cos 150t Volt !

Asumsikan pada t = 0, muatan pada kapasitor -0,05 C.

Penyelesaian :

Hukum kirchoff : $R\frac{dq}{dt} + \frac{1}{C}q = V$

Karena $R = 10, C = 4 \times 10^{-3}$ dan V = 85 cos 150t, diperoleh

$$10\frac{dq}{dt} + \frac{1}{4\cdot 10^{-3}}q = 85\cos 150t,$$ ~~atau bentuk~~

$$\frac{dq}{dt} + 25q = 8{,}5\cos 150t$$

P = 25

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C. 1) diketahui : $L = 2$, $R = 5\Omega$, $E = 12\,V$

a. Persamaan diferensial

$$L\frac{di}{dt} + Ri = E$$

$$2\frac{di}{dt} + 5i = 12$$

$$\frac{di}{dt} + \frac{5}{2}i = 6$$

b. solusi /output $i(t)$ bila $i(t=0) = 0$

Faktor integrasi $M = e^{\int 5/2\,dt} = e^{5/2t}$

$$i(t) = e^{-\int 5/2\,dt}\left(\int E e^{\int 5/2\,dt}\,dt\right)$$

$$= e^{-5/2t}\left(\int 6e^{5/2t}\,dt\right)$$

$$= e^{-5/2t}\left(6\int e^{5/2t}\,dt\right)$$

$$= e^{-5/2t}\left(6\cdot\frac{2}{5}e^{5/2t} + c\right)$$

$$= e^{-5/2t}\left(\frac{12}{5}e^{5/2t} + c\right)$$

$$= \frac{12}{5} + ce^{-5/2t} \rightarrow \text{solusi umum}$$

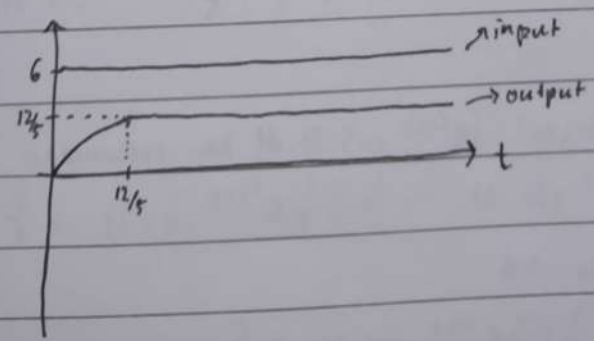
bila $i(t=0) = 0 \rightarrow$ $0 = \frac{12}{5} + ce^{-5/2(0)}$

$$0 = \frac{12}{5} + c$$

$$c = -\frac{12}{5}$$

$$i(t) = \frac{12}{5} - \frac{12}{5}e^{-5/2t} \rightarrow \text{solusi khusus}$$

c. Gambar grafik input dan output



2. Diketahui : $R = 5$, $L = 2$, $E = 12 \sin 3t$ V

a. Persamaan diferensial

$L\frac{di}{dt} + Ri = E$

$2\frac{di}{dt} + 5i = 12 \sin 3t$

$\frac{di}{dt} + \frac{5}{2}i = 6 \sin 3t$

b. Solusi /output bila $i(t=0) = 0$

$i(t) = e^{-\int R/L\, dt} \left(\int E e^{\int R/L\, dt}\, dt \right)$

$= e^{-\int 5/2\, dt} \left(\int 6 \sin 3t \cdot e^{5/2\, dt}\, dt \right)$

$i(t) = e^{-5/2t} \left(6 \int e^{5/2t} \sin 3t\, dt \right)$

* hitung integral $\int e^{5/2t} \sin 3t\, dt$

$u = \sin 3t \quad dv = e^{5/2t}\, dt \quad du = 3 \cos 3t\, dt \quad v = \frac{2}{5}e^{5/2t}$

$\int dvu = vu - \int vdu$

$\int e^{5/2t} \sin 3t\, dt = \frac{2}{5}e^{5/2t} \sin 3t - \int \frac{2}{5}e^{5/2t} (3 \cos 3t\, dt)$

$= \frac{2}{5}e^{5/2t} \sin 3t - \frac{3}{5} \int e^{5/2t} \cos 3t\, dt$

* hitung integral $\int e^{5/2t} \cos 3t\, dt$

$u = \cos 3t \quad dv = e^{5/2t} \quad du = -3 \sin 3t\, dt \quad v = \frac{2}{5}e^{5/2t}$

$\int dvu = vu - \int vdu$

$\int e^{5/2t} \cos 3t\, dt = \frac{2}{5}e^{5/2t} \cos 3t - \int \frac{2}{5}e^{5/2t} (-3 \sin 3t\, dt)$

$= \frac{2}{5}e^{5/2t} \cos 3t + \frac{3}{5} \int e^{5/2t} \sin 3t\, dt$

* substitusi hasil integral $\int e^{5/2t} \cos 3t\, dt$ ke persamaan $\int e^{5/2t} \sin 3t\, dt$

$\int e^{5/2t} \sin 3t\, dt = \frac{2}{5}e^{5/2t} \sin 3t - \frac{3}{5} \left(\frac{2}{5}e^{5/2t} \cos 3t + \frac{3}{5} \int e^{5/2t} \sin 3t\, dt \right.$

Misalkan $\int e^{5/2t} \sin 3t\, dt = A$

$A = \frac{2}{5}e^{5/2t} \sin 3t - \frac{3}{5} \left(\frac{2}{5}e^{5/2t} \cos 3t + \frac{3}{5}A \right.$

$A = \frac{2}{5}e^{5/2t} \sin 3t - \frac{6}{25}e^{5/2t} \cos 3t - \frac{9}{25}A$

$A + \frac{9}{25}A = \frac{2}{5}e^{5/2t} \sin 3t - \frac{6}{25}e^{5/2t} \cos 3t$

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$\frac{34}{25}A = \frac{2}{5}e^{5/2t}\sin 3t - \frac{6}{25}e^{5/2t}\cos 3t$

$A = \frac{25}{34}\left(\frac{2}{5}e^{5/2t}\sin 3t - \frac{6}{25}e^{5/2t}\cos 3t\right)$

$A = \frac{10}{34}e^{5/2t}\sin 3t - \frac{6}{34}e^{5/2t}\cos 3t)$

* substitusi A ke persamaan i(t)

$i(t) = e^{-5/2t}\left(6\left(\frac{10}{34}e^{5/2t}\sin 3t - \frac{6}{34}e^{5/2t}\cos 3t + c\right)\right)$

$= e^{-5/2t}\left(\frac{60}{34}e^{5/2t}\sin 3t - \frac{36}{34}e^{5/2t}\cos 3t + c\right)$

$i(t) = \frac{60}{34}\sin 3t - \frac{36}{34}\cos 3t + ce^{-5/2t}$

bila $i(t=0) = 0 \rightarrow 0 = \frac{60}{34}\sin 3(0) - \frac{36}{34}\cos 3(0) + ce^{-5/2(0)}$

$0 = 0 - \frac{36}{34} + c$

$c = \frac{36}{34}$

$i(t) = \frac{60}{34}\sin 3t - \frac{36}{34}\cos 3t + \frac{36}{34}e^{-5/2t}$ → solusi khusus

c. Grafik input dan output

